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MAGNETIC FIELD MODELLING AND SYMPLECTIC INTEGRATION OF MAGNETIC FIELDS ON CURVED REFERENCE FRAMES FOR IMPROVED SYNCHROTRON DESIGN: FIRST STEPS *

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Abstract

Compact synchrotrons, such as those proposed for cancer therapy, use short and highly bent dipoles. Large curvature drives non-linear effects in both body and fringe fields, which may be critical to obtain the desired dynamic aperture. Similarly to current practice for a straight magnet, our long-term goal is finding a parametrization of the field map to capture the relevant dynamical effects. This parametrization will then be used to optimize the performance of the synchrotron by multi-turn tracking simulations and, at the same time, drive the development of the magnet design by providing measurable quantities that can be computed from field maps. This paper presents the first steps towards the goal of representing the field with a few key parameters.

INTRODUCTION

To parametrize the short and highly bent magnets for cancer therapy [1], we start from a Taylor expansion of static magnetic fields in vacuum using Frenet-Serret coordinates and without assuming special symmetries. This Taylor expansion tends to the usual multipole expansion for a purely transverse field in a straight coordinate system. The first terms of the Taylor expansion are extracted from a field map and associated to beam dynamics observables. We present first results on two field maps, such as the canted dipole proposed for HITRIplus carbon-ion synchrotron [2] and the dipoles for the Extra Low Energy Antiproton Ring (ELENA) [3] as representative of traditional resistive magnets. We conclude by discussing the next steps.

TAYLOR EXPANSION

We use the Serret-Frenet coordinates to express the fields,

$$\vec{R}(x, y, s) = \vec{R}_0(s) + x\hat{x}(s) + y\hat{y}(s) \tag{1}$$

$$\frac{d}{ds}\vec{R}_0(s) = \hat{s}(s) \;, \tag{2}$$

where $\hat{s} = \vec{R}_0'(s)$, $\hat{x}'(s) = h(s)\hat{s}(s)$ and $\hat{y}(s) = -\hat{x}(s) \times \hat{s}(s)$ and $h(s) = 1/\rho(s)$ is the local curvature, see Figure 1 for the specific case that will be used later. The Laplacian of the

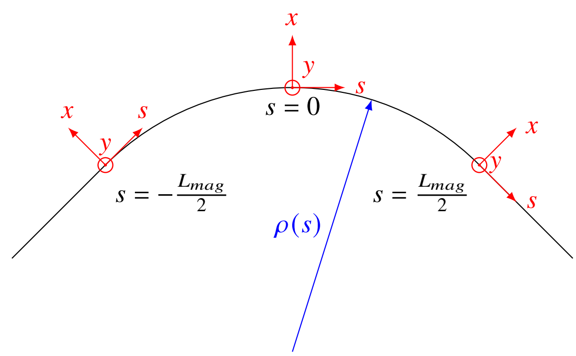

Figure 1: Frenet-Serret coordinate system used for the HITRIPlus canted-cosine-theta and Elena magnets.

magnetic scalar potential in the Frenet-Serret coordinates is given by the following expression,

$$\nabla^2\phi = \frac{1}{1+hx}\partial_x\left((1+hx)\partial_x\phi\right) + \partial_y^2\phi + \frac{1}{1+hx}\partial_s\left(\frac{1}{1+hx}\partial_s\phi\right) \;. \tag{3}$$

We can now write the scalar potential in our coordinate frame as a general expansion in y ,

$$\phi(x, y, s) = \sum_{i=0}^{\infty}\phi_i(x, s)\frac{y^i}{i!} \;, \tag{4}$$

where the functions $\phi_i(x, s)$ still have to be determined [1]. From the Maxwell equations, $\nabla^2\phi = 0$, one finds that ϕ_i have to satisfy the recurrence relation

$$\phi_{i+2} = -\frac{1}{1+hx}\left(\partial_x\left((1+hx)\partial_x\phi_i\right) + \partial_s\left(\frac{1}{1+hx}\partial_s\phi_i\right)\right) \;. \tag{5}$$

From this recurrence relation, only two initial functions can be independently chosen. These two functions, ϕ_0 and ϕ_1,

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[1] We use the approach in Ref. [4], rather than the traditional approach in Ref. [5], but another notation inspired by Ref. [6].

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can be expanded in x as follows,

$$\phi_0(x,s) = -a_0(s) - \sum_{n=1}^{\infty} a_n(s)\frac{x^n}{n!} \tag{6}$$

$$\phi_1(x,s) = -\sum_{n=1}^{\infty} b_n(s)\frac{x^{n-1}}{(n-1)!} \tag{7}$$

and we also define $b_s(s) = a_0'(s)$ for convenience.

The magnetic field can be derived from the potential $\vec{B} = -\nabla\Phi$ in curvilinear coordinates. The transverse field on the midplane and the longitudinal field on axis is given by

$$B_x(x, y=0, s) = -\partial_x\phi_0(x,s) = \sum_{n=1}^{\infty} a_n(s)\frac{x^{n-1}}{(n-1)!} \tag{8}$$

$$B_y(x, y=0, s) = -\phi_1(x,s) = \sum_{n=1}^{\infty} b_n(s)\frac{x^{n-1}}{(n-1)!} \tag{9}$$

$$B_s(x=0, y=0, s) = -\frac{1}{1+hx}\partial_s\phi_0(x,s)\Big|_{x=0} = b_s(s)\ , \tag{10}$$

where the prime indicates the derivative with respect to s, which allows identifying the coefficients a_n and b_n for $n > 0$ with the following derivatives on axis:

$$a_n(s) = \partial_x^{n-1}B_x(x,y,s)\big|_{x=y=0} \tag{11}$$

$$b_n(s) = \partial_x^{n-1}B_y(x,y,s)\big|_{x=y=0}\ . \tag{12}$$

Table 1 shows the first terms of the field expansion, from which one can recognize that the coefficients a_n, b_n fall back to the usual multipole expansion for straight magnets with transverse-only fields. We also included the term with $a_1(s)$ that were not present in Ref. [6]. The extra terms in the transverse fields that depend on h and the derivatives of B_s introduce extra coupling and non-linear effects that need to be carefully evaluated and will be part of future studies.

FIELD MAPS

The analysis will focus on two case studies: the dipole proposed for the HITRIPlus project [7], which is a superconducting canted-cosine-theta (CCT) magnet [2], and the dipole used in the ELENA ring, which is a standard resistive magnet. Table 2 lists the principal magnet parameters. Figures 2 and 3 show a top view of the geometry of the B_y field. The two dipoles have qualitative different fringe field regions, which could require different approaches for their modelling in tracking codes for optics and long-term stability studies.

PARAMETER FITTING

The first step of the analysis consists in extracting the magnetic field in the Frenet-Serret coordinates and estimating the coefficients $a_n(s)$, $b_n(s)$. A piecewise constant curvature reference curve has been used to start the analysis on the CCT magnets (see Figure 1).

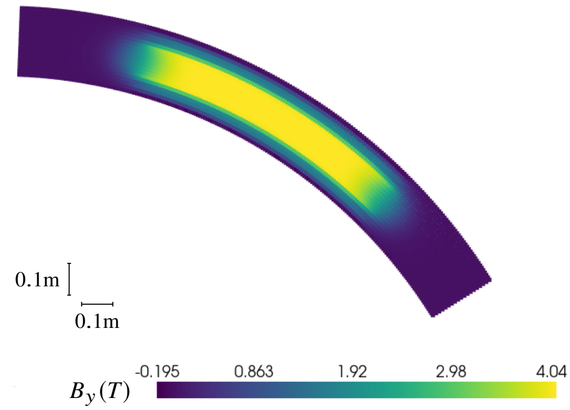

Figure 2: Vertical field map of the HITRIPlus CCT magnet in the midplane $y = 0$ of the magnet with magnetic length of 0.868 m.

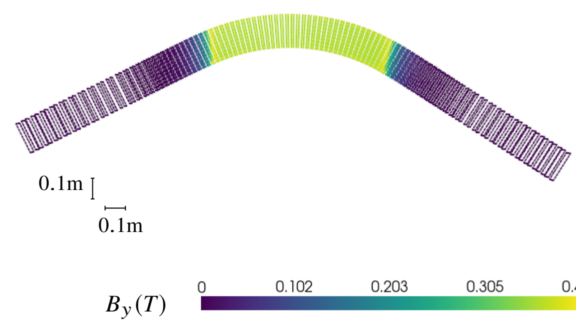

Figure 3: Vertical field map of the Elena resistive magnet [3] around the center of the aperture with magnetic length of 0.970 m.

Figure 4 shows the components B_y and B_s on axis and the determination of the magnetic length. Figures 5 and 6 show transverse fields as a function of x at a section of the magnet. The area inside the aperture can be well fitted by both a polynomial fit and by deriving the polynomial coefficients from Eqs. (11) and (12) using the central finite difference. This is a first step to obtaining a_n, b_n as a function of s.

CONCLUSION AND NEXT STEP

The next step consists of extracting the $a_n(s), b_n(s)$ coefficients and choosing how to model the s-dependences. After this step the magnet will have a fully analytical model inside the aperture, and we will be able to evaluate how many terms should be retained in order to obtain a good accuracy. We will then also be in a position to evaluate the advantages and disadvantages of a reference frame with piece-wise constant

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Table 1: First Terms of the Expansion of the Magnetic Field

	1	$\boldsymbol{x}$	$\boldsymbol{y}$	$\boldsymbol{x^2}$	$\boldsymbol{xy}$	$\boldsymbol{y^2}$
B_x	a_1	a_2	b_2	$\frac{a_3}{2}$	b_3	$\frac{-a_3-h(a_2-a_1h-2b'_s)+b_sh'-a''_1}{2}$
B_y	b_1	b_2	$-b'_s-a_1h-a_2$	$\frac{b_3}{2}$	$-a_3-h(a_2-a_1h-2b'_s)+b_sh'-a''_1$	$-\frac{b_3+hb_2+b''_1}{2}$
B_s	b_s	$-b_sh+a'_1$	b'_1	$b_sh^2-a'_1h+\frac{a'_2}{2}$	$-hb'_1+b'_2$	$-\frac{a_1h'+ha'_1+b''_s+a'_2}{2}$

Table 2: Magnet Parameters

Parameters	**HITRIPlus**	**ELENA**
Magnet type	AG-CCT	Resistive
Magnetic field	4 T	0.4 T
Magnetic length	0.868 m	0.970 m
Bending radius	1.65 m	0.927 m
Bending angle	30°	60°
Aperture	80 mm	76 mm

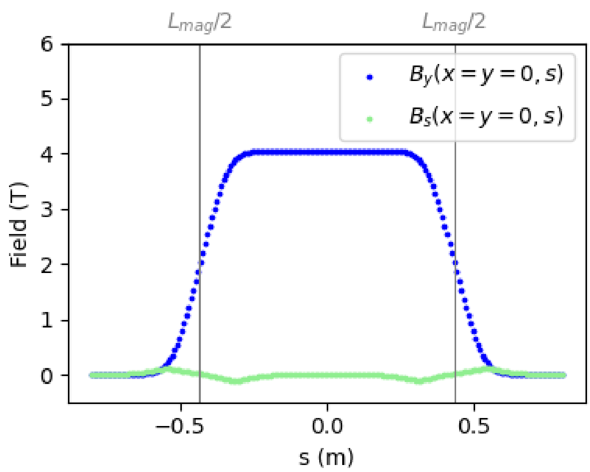

Figure 4: Magnetic field B_y and B_s of the HITRIPlus CCT magnet versus s direction.

h compared to one with a smooth $h(s)$ that follows more closely the particle trajectory. The analytical model allows a direct evaluation of the impact of the extra terms induced by the fringe-field and the curvature on the beam dynamics observables for a synchrotron (coupling, chromaticity, chromatic coupling, detuning with amplitude). In addition, we can evaluate how well the widely used SLAC-75 [8] PTC based fringe-field [9, 10] maps capture the correct physics, and can develop new maps if needed. It will also be beneficial to implement a symplectic integrator in Xsuite [11] that depends on the Taylor expansion terms. We plan to use the Wu-Forest-Robin [12] integrator and the Tao explicit integrator [13].

We will build simple synchrotron models with parameters suitable for cancer therapy using the dipoles studied. We will evaluate the importance of the non-linear components of the Taylor expansion, alone and together, to discriminate the most critical ones. By iteratively refining the parameters,

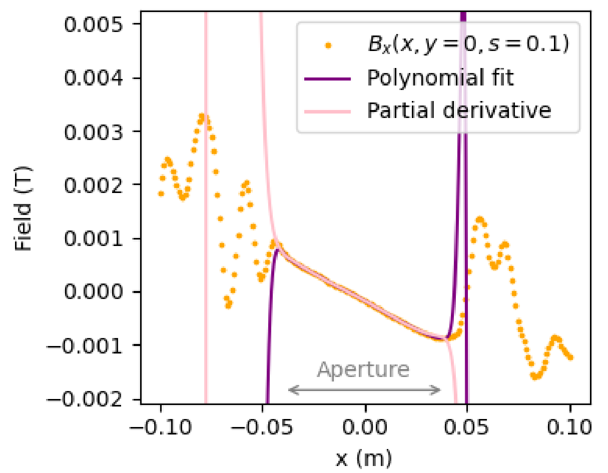

Figure 5: 20^{th} order fit of the B_x-field, both by direct polynomial fitting and by using partial derivatives, as described in Eqs. (11) and (12).

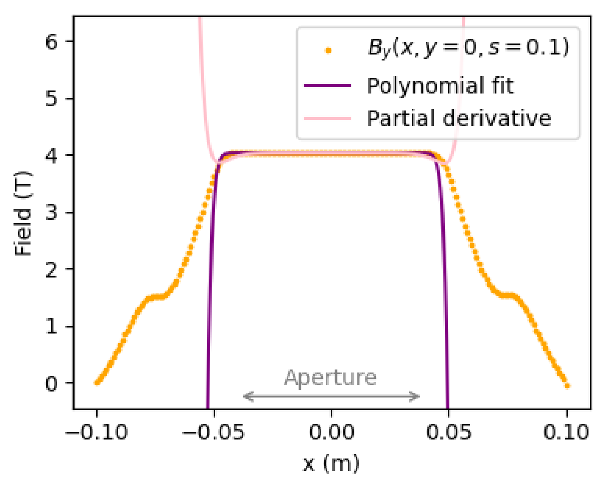

Figure 6: 20^{th} order fit of the B_y-field, both by direct polynomial fitting and by using partial derivatives, as described in Eqs. (11) and (12).

the objective is to achieve enhanced beam dynamics and extended synchrotron lifetime.

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